

Week 3 Assignment

1. 5 example problems for each :-

Supervised Learning:

- a. Agricultural Yield Prediction: Yield depends on measurable environmental and chemical inputs. So linear regression will be helpful the dependency of yield on these factors. Best is to use LR since this mathematically lies in that problem of dependency on factors.
- b. Water Boiling Time Prediction: It also depends on various factors room temp., heating temp., Vol. of water. Also a basic example of Linear Regression. Because it is highly predictable and purely mathematically consistent, the physics of it sits almost perfectly in a straight line.
- c. Exam Score Prediction: A simple example of predicting a student's final test score (out of 100) based strictly on the number of hours they spent studying. Best to use is SLR. $y = mx + c$ where m is the slope/weight of x which is number of hours the student study and c is the bias.
- d. Ad click problem: Predicting whether an online user will click banner ad(Yes/No) depending on time they spent on website. Helps in identifying the time threshold after which the ad should be shown to attract users. Since the output is just a probability of Yes or No(Binary) thus Logistic Regression is best to use here.
- e. Stock Prediction: Because a stock price is a continuous fluid number. We use MLR for predicting the stock price. As there are so many stock values factors affecting the predicted output.

Unsupervised Learning:

- a. Files Management: A bunch of different files in downloads folder they can be clustered into specific folders based on to their file types. Best is to use K-Means Clustering, it automatically structures messy data without needing to define rules for every single file.

- b. Library Book Sorting: Books being sorted into different sections based on their thickness and length. Best is to use Hierarchical Clustering. Because this helps in making a nested layout of the books that's how a library is in a structured manner. And we don't need to give number of clusters if using K means.
- c. T-shirt Organizing: A bunch of different sizes tshirts are given, we need classify them into 3 classes S,M&L. So best is to use K-means clustering as we know how many types of clusters we need i.e. 3. So it will easily identify the sizes based on the dimensions data.
- d. Color Palette Extractor: An image is given we need to extract the color palette based on that image getting the 3 dominant colors. So K-means clustering will be the best here as we again know the number of clusters that we want and the model will check on the data of each pixel's rgb values.
- e. Lifestyle Classifier: People come from different backgrounds socially and economically. So such social and economic factors can be used to find similarities and group them into certain clusters / classes. This can be easily done using K-means clustering as it helps in identifying these similarities and grouping into clusters based on that. This can be helpful for identifying the user and his/her preferences and provide services/products accordingly.

2. Stock Values To Apply Regression

Stock Values - Continuous numerical data points tracked over time. Regression is perfect for such scenarios to predict a continuous number

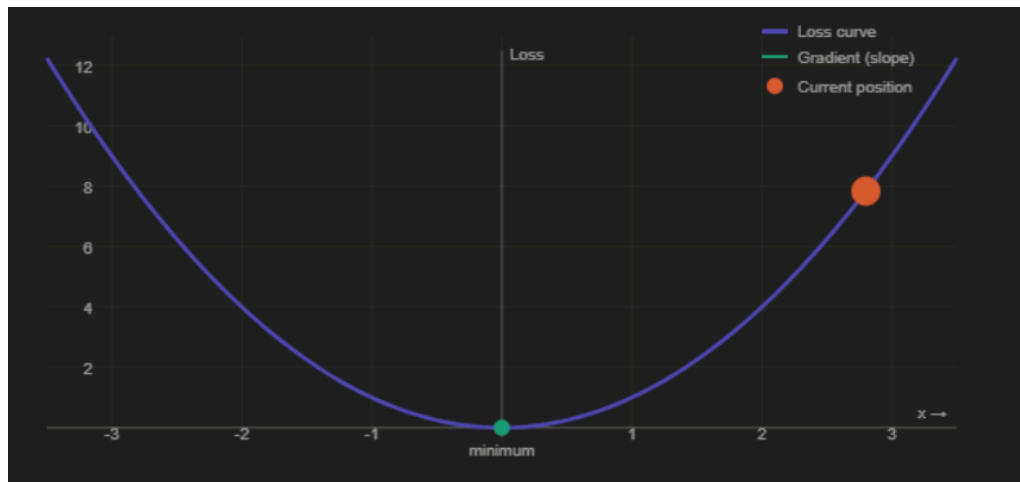
E.g. Previous closing opening high low values of the stock, ongoing prices, trading volume, moving average values, market indices, etc.

SLR won't be fine to predict stock value as it is clear from the above different types of stock values that the prediction will depend on such so many factors. Thus MLR is the best for such case.

The feature space for MLR should include mix of different market drivers:-

Such as Previous Day's Closing price, Trading Volume, Moving Averages, Market Index Performance, etc.

4. Gradient Descent is simply a mathematical tool to find the weights & biases such that the cost function is minimized which is MSE in Linear Regression and Log Loss in Logistic Regression for max accuracy.



So this is a simple graph example depicting graph descent so the steps of graph descent are first start from any point on the curve then determine where to go to reach minima by computing the gradient at that point and going -ve of that direction and move there by $-\nabla C$ repeat until reached the global minima.

5. Backpropagation is an algorithm for computing this overall crazy complicated gradient it is used in complex systems where changing a parameter at the start alters the structural integrity at the very end.

Imagine a civil engineer designing a steel bridge. They want to minimize the structural strain at the centre of the bridge when a heavy truck drives over it.

So here backpropagation is applied since bridge is modeled as a mesh grid of connected physical nodes(a structural graph). Final outcome is deformation at the center. To optimize the bridge design they use reverse automatic differentiation (mathematical definition of backpropagation). Calculating the error (too much strain) at the final output and passing that calculation backward through the connection links using the calculus Chain Rule.

This tells exactly how changing the thickness or material of an internal support pillar at the beginning will reduce strain at end.